

ARUNIT BAIDYA

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Master of Science, Computer Science

Aug 2024 – April 2026

Boston, MA, US

Anna University, PSG College Of Technology

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 – July 2022

Coimbatore, India

WORK EXPERIENCE

Arcesium | *Data, Backend, Cloud-Native*

Software Engineer

Jul 2022 - Jul 2024, Bangalore

- Profiled and optimized **Python/Java code** in distributed big data processing pipelines, **improving process times by 40%**.
- Built **JSON** config-driven data-filtering module for Spark ETL in **reducing compute usage by 30%** via partition pruning.
- Developed event-driven microservices with AWS SNS/SQS, **reducing latency by 25% through** batch size and timeout tuning.
- Optimized resource provisioning sizes (T-shirt model) in **ArgoWorkflow** and **Kubernetes**, **adopted in 10+ workflows**.
- Automated CI/CD pipelines with **Jenkins, Docker, & Kubernetes**, integrating JUnit for *testing ensuring 85% coverage*.
- Created LLD documentation with business logic translation, **reducing** cross-functional team **inquiries by 50%**.
- Actively participated in **Agile/SCRUM** teams to deliver **iterative enhancements** and optimize project workflows.

Software Engineering Intern

Feb 2022 - May 2022, Remote

- **Increased Spark ETL throughput by 70%** via preprocessing stage to split heterogeneous data for parallel processing.
- Enhanced system observability by integrating log publishing to **Elasticsearch** and creating **Datadog** monitors.

Software Engineering Intern

May 2021 - Jul 2021, Remote

- Automated PDF Parsing in **python** to identify and parse tabular information using OpenCV and Camelot.
- Implemented **CRON processing** to parse PDFs mapped against filters & annotations, **saving 4 hours work/week**.

PROJECTS

Scalable Image Sharing Platform ([GitHub](#)) | *Golang, JavaScript, Kafka, Redis, Postgres, MinIO/S3*

- Architected a high-performance Golang backend utilizing **PostgreSQL** and **MinIO/S3** for scalable, reliable storage.
- Developed **RESTful APIs** for image uploads, likes, dislikes, views, and comments with **batched database updates**.
- Leveraged **Kafka** for event-driven processing and **Redis** caching (TTL / LRU eviction) improving request response times by **40%**.
- Deployed using **Docker**, with **JWT authentication, rate limiting, & Prometheus** monitoring for enhanced observability.

Image Processing Application ([GitHub](#)) | *Java, SOLID, Object-oriented, MVC*

- Developed an image processing application in **Java** supporting 10+ operations, including blur, compress, and downscale, with execution options via terminal, script, or GUI inputs.
- Used SOLID principles, MVC architecture, and **design patterns** (Command, Factory etc.), **reducing code complexity by 40%**.

Scalable TODO-List API ([GitHub](#)) | *Golang, Kafka, Redis, Docker, Ansible*

- Designed a scalable REST API for a TODO-List app with **PostgreSQL**, leveraging Redis caching, paginated list queries, event-driven architecture, and automated deployments via **Ansible** and **GitHub CI/CD**, handling up to 20,000 queries per second.

egOS ([GitHub](#)) | *C, Operating Systems, RISC V, Multithreaded Programming*

- Enhanced egOS by implementing **multilevel feedback queue** (MLFQ) kernel scheduling, **user-level threading, virtual memory management** via paging, and **physical memory protection** on RISC-V architecture in **C**.

SKILLS

Programming Languages: Python, Java, Go, C, JavaScript, HTML, CSS

Frameworks and Libraries: REST, Redis, Swing, Pytest, JUnit, Argo Workflow, Ansible, Prometheus

Data Engineering: Spark, Kafka, ETL, DeltaLake, Snowflake, PostgresDB, MySQL, NoSQL, RDBMS

Cloud and DevOps: REST API development, AWS, Elasticsearch, SNS, SQS, Terraform, Docker, Kubernetes

Software Development: Data Structures & Algorithms, Agile Development, Test-Driven Development, Code Reviews, Git